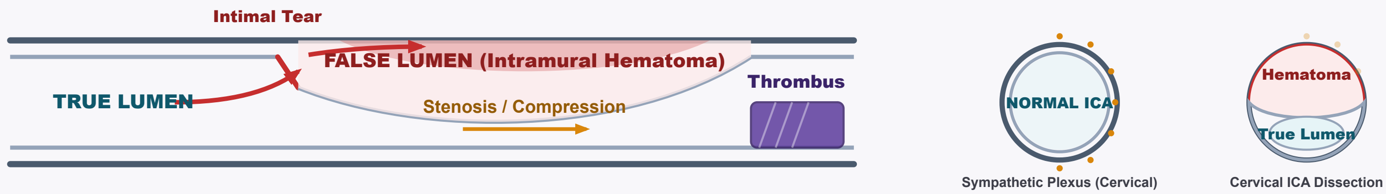
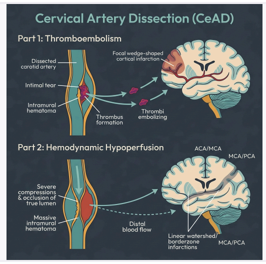
# Cervical Artery Dissection




## 1. Clinical Presentation & Pathophysiology

### Ipsilateral Pain & Onset

- **Carotid (ICA)**: Frontotemporal/retro-orbital/facial pain (jaw angle).
- **Vertebral (VA)**: Severe occipital or posterior neck pain.
- **Onset**: Precedes stroke/TIA by hours to days (median 4 days).

### Anhidrosis-Sparing Horner's

- **Signs**: Ptosis/miosis (28–58% of ICA) **without** anhidrosis.
- **Mechanism**: Sweat fibers follow ECA plexus; pupil/eyelid fibers follow ICA.

### Neurological Deficits

- **CN Palsies**: CN IX–XII palsies (8–16%) from local ICA compression.
- **VA Territory**: Wallenberg syndrome, cerebellar ataxia, PICA/AICA strokes.

## 2. Diagnostic Workup

- **CTA Head/Neck**: Shows string sign, dissection flap, pseudoaneurysm, or occlusion.
- **MRI Neck (T1 Fat-Sat)**: Pathognomonic crescent sign (intramural hematoma).
- **DSA**: Reserve for diagnostic doubt or stenting.
- **Screening**: Assess for FMD/connective tissue disease, especially if spontaneous/recurrent.

## 3. Medical Management: Extracranial vs. Intracranial Dissection

### Extracranial Dissection

- **Antithrombotics**: Continue for at least 3–6 months (Class I, ESO/AHA).
- **Choice**: SAPT vs VKA/DOAC; short-term DAPT (21–90d) is reasonable (ESO consensus).
- **STOP-CAD**: In occlusions, consider anticoagulation Day 1–30, then switch to antiplatelet.
- **IV Thrombolysis**: Safe & indicated within 4.5 hours (Class I).

### Intracranial & Pseudoaneurysms

- **SAH**: Lack external elastic lamina & thin adventitia; rupture risk.
- **Anticoagulation**: Avoided if SAH present. Prefer single antiplatelet.
- **IVT Caution**: IVT is safe in extracranial CeAD but safety/efficacy in intracranial extension is not well established (AHA 2024).
- **Stenting**: Reserve for recurrent ischemia despite optimal medical therapy or severe flow-limiting stenosis.

- **Recurrence & Activity**: Long-term CeAD recurrence is low (~1%/yr). Avoid high-risk neck activities (chiropractic neck manipulation, rollercoasters, extreme hyperextension/rotation) for secondary prevention (AHA 2024 / ESO 2021).

## 4. Landmark Trial & Cohort Evidence

| Study / Year | Population & Design | Interventions Compared | Key Outcomes & Clinical Nuance |
|---|---|---|---|
| **CADISS** 2015 | N = 250. Extracranial CeAD. RCT. | Antiplatelet vs. Anticoagulant for 3 months. | • **Primary Composite (Ipsilateral stroke or death at 3m)**: 2.0% vs. 1.0% (p = 0.63). Established clinical equipoise. |
| **TREAT-CAD** 2021 | N = 194 (PP = 173). Extracranial. RCT. | Aspirin 300mg daily vs. VKA for 3 months. | • **Primary Composite (Stroke, major hemorrhage, death, or new MRI lesion at 14d)**: 23% vs. 15% (Non-inferiority NOT met). Ischemic stroke: 8.0% vs. 0%. |
| **Kaufmann IPD** 2024 | N = 444. IPD meta-analysis of CADISS + TREAT-CAD. | Antiplatelet vs. Anticoagulant x 90d. | • **Primary Composite (Ischemic stroke, major bleeding, or death at 90d)**: No significant difference (1.4% anticoagulation vs. 4.4% antiplatelet, p = 0.11).<br>• **Ischemic Stroke alone**: Significant reduction with anticoagulation (0.5% vs. 4.0%; OR 0.14, p = 0.01), with a non-significant increase in major bleeding (0.9% vs. 0%). |
| **STOP-CAD** 2024 | N = 3,636. Multicenter observational cohort registry. | Antiplatelet vs. Anticoagulation. | • **Temporal Risk**: 87% of recurrent strokes occurred in the first 30 days.<br>• **Occlusion Benefit**: Patients with complete arterial occlusion at baseline had the highest stroke risk and benefited most from early anticoagulation.<br>• **Transition Strategy**: Initiate anticoagulation for days 1–30 (highest stroke risk window) and then transition to antiplatelet monotherapy at day 30 to mitigate long-term bleeding risks (which rise significantly by day 180). |


**CADISS:** *Lancet Neurol*. 2015;14(4):361-7. PMID: 25684164 | **TREAT-CAD:** *Lancet Neurol*. 2021;20(5):341-350. PMID: 33765420
**Kaufmann IPD:** *JAMA Neurol*. 2024;81(6):630-637. PMID: 38739383 | **STOP-CAD:** *Stroke*. 2024;55(4):908-918. PMID: 38334460 | **AHA/ASA:** *Stroke*. 2021;52:e364-e467. PMID: 34024117 | **AHA Statement 2024:** *Stroke*. 2024;55:e84-e107. PMID: 38301552 | **ESO Guideline 2021:** *Eur Stroke J*. 2021;6(3):XXXIX-LXXXVIII. PMID: 34528453